

Raj Vaibhav

LinkedIn: <https://www.linkedin.com/in/raj-vaibhav>

Github: <https://github.com/Raj-Vaibhav>

Email: rajvaibhav003@gmail.com

Mobile: 8287224859

SKILLS

Languages:	C++, C, JavaScript, Python, Java
Backend/Web:	Node.js, Express.js, REST APIs, HTTP, JSON, PostgreSQL
Systems/Tools:	POSIX Sockets, Multithreading, Linux, Git
Soft Skills:	Adaptable, Quick Learner, Problem-Solver

PROJECTS

File Compression Tool | [GitHub](#)

Jan' 26

- Engineered a binary-safe **lossless** file compression tool in C++ using **Huffman Coding**, supporting compression and decompression of arbitrary file types (text, PDF, binary) via a custom **CLI**.
- Designed a **custom compressed file format** with metadata headers for deterministic tree reconstruction, implementing bit-level encoding/decoding using STL, priority queues, and manual bit packing.
- Benchmarked compression performance on real-world datasets (up to **100MB+ files**), achieving **~67%** reduction on repetitive data and **~39%** on natural language text while ensuring exact **byte-level restoration**.
- Tech Stack:** C++, STL, Bit Encoding, Priority Queue, Huffman Tree

Mini HTTP Server | [GitHub](#)

Nov' 25

- Built a **multithreaded HTTP server** in C++ from scratch using **POSIX TCP sockets**, implementing the HTTP request/response lifecycle and **OS-level concurrency**.
- Designed and implemented a custom **thread pool** (using `std::thread`, `mutex`, and `condition_variable`) to handle **concurrent client connections** efficiently, supporting dynamic routing and static file serving.
- Benchmarked the server using **ApacheBench** (~10k+ req/sec locally) and analyzed latency, **concurrency scaling**, and CPU utilization to evaluate performance under load.
- Tech Stack:** C++, POSIX Sockets, Thread Pool, HTTP, TCP/IP

TRAINING

CSE Pathshala

Jun' 25 – Jul' 25

- C++ Programming: OOPs and Data Structures
- Completed **35+ hours** of intensive live summer training focused on Object-Oriented Programming, Core C++, and DSA fundamentals.
- Practiced **implementation of key data structures** (arrays, linked lists, stacks, queues, trees, graphs) and **OOP principles** (classes, objects, inheritance, polymorphism, abstraction, encapsulation).
- Attained **problem-solving** proficiency by completing structured C++ assignments and passing a final assessment.

CERTIFICATES

C++ Programming: OOPS and DSA | *CSE Pathshala*

Aug' 25

Fundamentals of Network Communication | *Coursera*

Sept' 24

Introduction to Hardware and Operating Systems | *Coursera*

Sept' 24

ACHIEVEMENTS

- Accepted into **GirlScript Summer of Code (GSSoC) 2026**; selected to contribute to open source projects as part of a nationally recognized program.
- Solved 200+ algorithmic problems across **LeetCode**, **Hacker Rank** and **GeeksforGeeks** showcasing algorithmic thinking and strong data structures knowledge.

EDUCATION

Lovely Professional University

Bachelor of Technology
Computer Science and Engineering; CGPA: 7.32

Phagwara, Punjab
Aug' 23 – Aug' 27

Siddhartha Public School

Intermediate
PCM; Percentage: 73.6%

Kanjhawala, Delhi
Mar' 21 – May' 22

Rotary Public School

Matriculation
Percentage: 91%

Gurugram, Haryana
Apr' 19 – Jan' 20